

Jintao Li
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Zhejiang University, Hangzhou, Zhejiang, China

Experience

2025–
Present **Postdoctoral Researcher**, Zhejiang University
Advisor: Prof. Weichang Li

Education

2020–2025 **Ph.D.**, Geophysics, University of Science and Technology of China
Advisor: Prof. Xinming Wu
Thesis: *High-Resolution Post-Stack Seismic Data Processing and Applications Using Deep Learning*

2016–2020 **B.Sc.**, Geophysics, University of Science and Technology of China

Research Interests

- AI and Foundation Models for Geoscience
- Seismic Processing, Interpretation, and Inversion
- DAS, Fiber-Optic Sensing, Multi-Sensor Fusion, and Geophysical Monitoring
- Geoscience AI Infrastructure and Open-Source Software

Research Funding

2027–2029 **Young Scientists Fund (Category C), National Natural Science Foundation of China**
Principal Investigator; RMB 300,000; Jan. 2027–Dec. 2029
Formerly the Young Scientists Fund

2026–2028 **Fellowship from the China Postdoctoral Science Foundation**
Principal Investigator; Certificate No. 2026M790497; RMB 80,000; Sep. 2026–Aug. 2028

2026–2028 **Postdoctoral Fellowship Program (Grade B) of China Postdoctoral Science Foundation**
Principal Investigator; Grant No. GZB20260111

Publications

 [Google Scholar profile](#)

* → Equal contribution, † → corresponding author

Published Research

1. Wang, Yuting, **Jintao Li**[†], Xiaoming Sun, and Xinming Wu[†]. 2026. “Deep Learning-based Seismic Reflectivity Estimation by Pre-training on Labeled Synthetic Data and Physics-guided Fine-tuning in Field Data.” *Geophysical Journal International (GJI)*, ggag317. <https://doi.org/10.1093/gji/ggag317>.
2. **Li, Jintao**, Xinming Wu, Xianwen Zhang, Xin Du, Xiaoming Sun, Bao Deng, and Guangyu Wang. 2026. “High-Fidelity Seismic Super-Resolution Using Prior-Informed Deep Learning with 3D Awareness.” *IEEE Transactions on Image Processing (IEEE TIP)* 35:2152–2166. <https://doi.org/10.1109/TIP.2026.3663926>.
– **CCF-A, IF: 15.3.**
3. **Li, Jintao**, and Xinming Wu. 2026. “Memory-Efficient Full-Volume Inference for Large-Scale 3D Dense Prediction without Performance Degradation.” *Communications Engineering* 5:20. <https://doi.org/10.1038/s44172-025-00576-2>.
– **Nature Portfolio, IF: 7.0**
– **This paper was unanimously recognized by all three reviewers.**
4. **Li, Jintao**, Yunzhi Shi, and Xinming Wu. 2025. “CIGVis: An Open-source Python Tool for the Real-time Interactive Visualization of Multidimensional Geophysical Data.” *Geophysics* 90 (1): F1–F10. <https://doi.org/10.1190/geo2024-0041.1>.
5. **Li, Jintao**, Xinming Wu, Yueming Ye, Cun Yang, Zhanxuan Hu, Xiaoming Sun, and Tao Zhao. 2023. “Unsupervised Contrastive Learning for Seismic Facies Characterization.” *Geophysics* 88 (1): WA81–WA89. <https://doi.org/10.1190/geo2022-0148.1>.
6. **Li, Jintao**, Xinming Wu, and Zhanxuan Hu. 2022. “Deep Learning for Simultaneous Seismic Image Super-Resolution and Denoising.” *IEEE Transactions on Geoscience and Remote Sensing (IEEE TGRS)* 60:1–11. <https://doi.org/10.1109/TGRS.2021.3057857>.
– **ESI Highly Cited Paper, IF: 9.4.**
7. Gao, Hui, Xinming Wu, **Jintao Li**, Xiaoming Sun, and Jiarun Yang. 2026. “Massive-scale Unlabeled Field and Labeled Synthetic Seismic Datasets of Global Shelf-edge Clinothems.” *Geophysical Journal International (GJI)*, ggag295. <https://doi.org/10.1093/gji/ggag295>.
8. Sheng, Hanlin, Xinming Wu, Xu Si, **Jintao Li**, Sibao Zhang, and Xudong Duan. 2025. “Seismic Foundation Model: A Next Generation Deep-learning Model in Geophysics.” *Geophysics* 90 (2): IM59–IM79. <https://doi.org/10.1190/geo2024-0262.1>.
– **ESI Highly Cited Paper.**
9. Han, Long, Xinming Wu, Zhanxuan Hu, **Jintao Li**, and Huijing Fang. 2024. “MAMCL: Multi-attributes Masking Contrastive Learning for Explainable Seismic Facies Analysis.” *Computers & Geosciences* 193:105731.

Unpublished Research

1. **Li, Jintao**, Weichang Li, Kai Tong, and Xaingyu Guo. 2026. *FiLark: A Streaming-first Software Framework for End-to-end Exploration, Annotation, and Algorithm Integration in Distributed Acoustic Sensing*. arXiv: [2605.20132](https://arxiv.org/abs/2605.20132). <https://arxiv.org/abs/2605.20132>.
– Submitted to Computers & Geosciences; under review.
2. Han, Long, Xinming Wu[†], Xiaoming Sun, **Jintao Li**, and Yimin Dou. 2026. *A Universal Diffusion Regularizer for Geophysical Inversion*.
– Under review.
3. Dou, Yimin, Xinming Wu, Nathan L. Bangs, Harpreet Singh Sethi, **Jintao Li**, Hang Gao, and Zhixiang Guo. 2026. *Geological Everything Model 3D: A Promptable Foundation Model for Unified and Zero-shot Subsurface Understanding*. arXiv: [2507.00419](https://arxiv.org/abs/2507.00419). <https://arxiv.org/abs/2507.00419>.
– In revision.
4. Sheng, Hanlin, Xinming Wu, Hang Gao, Haibin Di, Sergey Fomel, **Jintao Li**, and Xu Si. 2025. *On the Workflow, Opportunities and Challenges of Developing Foundation Models in Geophysics*. arXiv: [2504.17384](https://arxiv.org/abs/2504.17384). <https://arxiv.org/abs/2504.17384>.

Other Conference Papers and Abstracts

1. **Li, Jintao**, Yunzhi Shi, and Xinming Wu. 2024. "CIGVis: An Open-source Python Tool for Real-time Interactive Visualization of Multidimensional Geophysical Data." In *85th EAGE Annual Conference & Exhibition*, 1–5. <https://doi.org/10.3997/2214-4609.2024101065>.
2. **Li, Jintao**, Xinming Wu, and Zhanxuan Hu. 2020. "Deep Learning for Simultaneous Seismic Super-Resolution and Denoising." In *SEG International Exposition and Annual Meeting*. <https://doi.org/10.1190/segam2020-3426412.1>.
3. Sheng, Hanlin, Xinming Wu, Xu Si, **Jintao Li**, Sibao Zhang, and Xudong Duan. 2024. "Seismic Foundation Model (SFM): All-purpose Feature Extraction from Seismic Data for Diverse Geophysical Applications." In *85th EAGE Annual Conference & Exhibition*, 1–5. <https://doi.org/10.3997/2214-4609.202410133>.
4. Si, Xu, Xinming Wu, **Jintao Li**, Xin Cui, Hanlin Sheng, Hang Gao, Zefeng Li, and Zhigang Peng. 2024. "Tuning into Earth's Voice: Leveraging Large Language Models for Earthquake Detection." In *AGU Fall Meeting Abstracts*, S11B–3384. <https://ui.adsabs.harvard.edu/abs/2024AGUFMS11B.3384S/abstract>.

Tools & Software

- [cigvis](#): A tool for visualizing multidimensional geophysical data.
- [cigsegy](#): A tool for exchanging data between SEG-Y format and NumPy array inside Python environment.
- [SeismicSuperResolution](#): A repository for the paper "Deep Learning for Simultaneous Seismic Image Super-Resolution and Denoising" (IEEE TGRS).

- [torchseis](#): A repository for the paper "Memory-Efficient Full-Volume Inference for Large-Scale 3D Dense Prediction without Performance Degradation" (Communications Engineering).

Awards & Honors

2025	Outstanding Graduate of University of Science and Technology of China (USTC)
2024	PetroChina Scholarship
2021	National Graduate Scholarship
2021–2025	First-Class Academic Scholarship, USTC
2020	Outstanding Undergraduate Thesis Award, USTC
2019	National Endeavor Scholarship

Teaching

University of Science and Technology of China (USTC)

2023	Invited Lecturer, <i>Artificial Intelligence in Geophysics</i> Delivered an independent 3-hour module on Python-based Scientific Data Visualization .
2021	Lead Teaching Assistant, <i>Artificial Intelligence in Geophysics</i> Coordinated 8 TAs; contributed to assignment design and grading rubrics.

Academic Service

Conference Service

Nov. 2026	Technical Committee Member , SEG-GeoAI 2026 Workshop, Hangzhou, China
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Journal Reviewer

Geophysics; IEEE TGRS; IEEE TSP; Sci. Rep.; Comput. Geosci.; JGR-MLC; IEEE GRSL; Interpretation; JGE.